

Forty Pounds Over

Implementation summary — *features/quest_the_bale_that_gained_forty_pounds/*

Quest 02 of features/systemic_quest_suggestions.md, promoted to spec 2026-08-30. Full design in README.md; ten worked routes in PLAYTHROUGHS.md. This document is only the part that has to be built.

What the quest is, in one paragraph

A corded bale of broadcloth weighs forty pounds more at the Wool Gate than it did on the Tallage beam two hours earlier. Officers call it smuggling, turn Hugh Crake's road cart around before a crowd, and impound the load until a public opening at the next Dayspring. Hugh is innocent; a crime did occur. Renn Crake, out of his depth on a salt forfeit and holding a promise he could not keep, diverted the cart through the bonded warehouse, cut the cord, added a weaver's off-book bolt and imitated the weigher's seal. The player has about a game day and a half to decide what the truth is worth, and to whom.

The one design constraint that governs the build

The discrepancy is arithmetic, not a flag. The bale's weight is the sum of what is physically inside it; the manifest is a separate written claim; the seal records what the beam called at sealing. Nothing anywhere stores “this bale is suspicious”. Remove the extra bolt and the bale genuinely weighs what its paper says. Add something else and the player has made it worse. This is what stops the quest being a dialogue tree with a stage, and it is why the resolution function reads no quest flags. Every engineering decision below follows from it — in particular, **there must never be an `is_tampered` boolean.**

What already exists (so the scope is smaller than it looks)

Nothing in the fiction has to be invented. Every named person is a shipped, authored character whose own sheet already contains their role; every location is a registered place with a baked nav pin; the cart is already in the data.

Already shipped	What it gives the quest
<code>rounds.json</code> → <code>road_parties[0]</code>	The <code>brede_wool_gate</code> party: Hugh Crake as leader, two carters, the Wool Gate, broadcloth as declared return cargo, and manifests that are already first-class objects in <code>round.rs</code> .
<code>notices.rs</code>	The full accusation ladder — Word → Summoned → Warranted — with accused, wronged and taken fields, <code>raise_notice</code> / <code>settle_notice</code> , and a summons that names a bell. The pressure on Hugh is entirely existing code.
<code>custody.rs</code>	Seizure, the escort tether, the Stone House, the posted fee and the fifth door out. Every route through arrest already works.
<code>items.json</code>	cloth with <code>grade</code> ∈ {kersey, broadcloth} at 14/40 sparks; key, wax, ledger, deed, letter, <code>tally_stick</code> , <code>writing_kit</code> . badge already carries an authenticity: [counterfeit] domain — the precedent for a forged seal.
18 character sheets	The weigher is going blind and has his daughter read the beam aloud; the carter knows every weight by the beam's note and cannot read; the notary is lying awake about a seal he failed to look twice at; the broker has already pledged his mother's warehouse keys. No sheet needs editing.
12 registered places	The Tallage, its toll-house, the bonded warehouse, the bonded weighing yard, the Tally Bridge, the Wool Gate, the Draper's Reach, the Needle, the Stone House. No nav rebake.

What has to be built

A. Three general capabilities (not quest-gated)

- Item weight** — an optional `weight_lb` on `items.json` kinds, keyed exactly like `price_sparks` so metadata variants differ (broadcloth 40 lb, kersey 24, wool bundle 28, grain sack 56). `ItemCatalog::weight_lb(&Item)`, multiplied by stack quantity. Absent means zero; only lots and beams ever ask.
- crates/cathedral-sim/src/lots.rs** — sealed, weighed consignments: real contents as item ids, a separate declared manifest, the weight the beam called at sealing, a seal impression, a cord state, and an append-only custody chain whose unrecorded leg is the crime's shape.
- The public weighing** — Ewart Tarn's spoken procedure as a sim event: the standard cited, the figure called, who read it aloud, and everyone inside the ordinary 20 m hearing radius when it was. Witnesses reuse the existing single authoritative recipient calculation.

These three are the reason the quest is worth building now rather than later: suggestions 13 (the fish cart), 14 (the locked grain store) and 16 (the disputed standard weight) all silently assume them, and become content problems once they exist.

A hard prerequisite that is not this quest's to build

features/knowledge_and_rumor/ must land first, through its own M4. It owns Fact/holds(), the what_you_know prompt block, rumour propagation and garbling, the player's receipts, the journal overlay, and World::arm_actor. Decided 2026-08-30 to ship both halves as one feature rather than an interim fact-only layer. The quest then authors JSON facts and owns no knowledge code at all — an earlier draft of this spec invented a per-quest knowledge enum, which is exactly the thing three quests each inventing one would have produced.

The schedule cost is real and should be weighed here: that feature's own principal risk is a perceptibility tuning pass — the LLM must voice an injected line often enough for the player to feel news travelling, without every mouth in a ward parroting the same sentence — and all three quest specs are now behind it.

```
impl Lot {
    /// Sum of the contents' catalog weight. Never stored, always computed.
    pub fn true_weight_lb(&self, world: &World) -> u32 { ... }

    /// What the opening will find. Positive means heavier than sealed.
    pub fn discrepancy_lb(&self, world: &World) -> i64 {
        self.true_weight_lb(world) as i64 - self.sealed_weight_lb as i64
    }
}
```

B. Quest-specific

- **BaleQuest state on World behind an absent/default-off data gate: phase machine (Dormant → Arming → Seized → Investigating → OpeningDue → Opened → Resolved), the clock, per-character standing, dispositions, outcome. Receipts and the player's casebook are NOT here** — they belong to the knowledge layer, because all three quests need them.
- **The Arming phase, one game day before the seizure: the bolt minted as a real item in the weaver's workshop, the promise seeded as an authored fact, Renn's goal and round set through World::arm_actor, Ede's errand actually walked. Arming is played, not narrated** — a player in the Wickmarket that day can watch it happen, so nobody needs to remember it.
- **The quest-day leg override on brede_wool_gate** — the shipped legs put the cart at the gate at Lamplight, which would leave the player only the night; on a quest day Hugh works ahead of his round to catch the Brede road, so the stop is at High Wick. Off when the quest is off.
- The public opening at the bonded warehouse: a pure function of lot state at that instant, reading no quest flags, with five branches on (contents vs declared, weight vs seal, seal authenticity, cord state).
- **Five outcome packages applied to the live world** — what the Draper's Reach pays by the ell, who brokers freight at the Tally Bridge, whether the Brede cart runs, queue length at the Tallage, and whether a real smuggling route with a copied key now exists.
- **One new data file, assets/world/quests/bale.json** — the lot, the seal, the cord, the custody chain, the outcome packages, and the quest's authored facts. Absent means the quest is off and golden prompts are byte-identical to today.

C. Seams

Three new player commands, all about acting on the bale: PlayerExamine (ask a competent nearby character to read a seal, a cord or a beam), PlayerStageWeighing and PlayerLodgeAmendment. One new message, EngineMessage::BaleQuest, on a dedicated monotonic quest revision.

There is no AcceptQuest and no offer scene. The two sibling quests are handed to the player by a named NPC with a writ; this one must not be, because Hugh Crake does not know he needs help and would not ask a stranger. The quest is walked into — by sight of the Wool Gate at High Wick, by earshot of a turned cart and frightened oxen, or by the word reaching the ward as ordinary garbled gossip. There is nothing to accept; what activates is the journal, on the first fact learned. The cost is that a player can miss it entirely, which is acceptable for a quest but would not be for a first one.

Do not put the journal in the actor/item PublicSnapshot or touch the public-state revision per fact learned — that republishes the whole cast and the configured crowd, and PublicSnapshot's 160 KiB bound already has little headroom. The Bevy side never re-derives weights, seal authenticity or guilt, and two fields are absolute: Lot::seal.pressed_by and FactSource — who pressed the wax, and why a fact is true — appear in no projection, prompt, log line, journal entry or HUD. A character who holds “the seal is an imitation” knows the seal is wrong; they do not know whose hand it was.

Milestones

Prerequisite: features/knowledge_and_rumor/ through its own M4. Nothing below starts before it.

	Milestone	Contents	Done when
M0	Weight, lots, one clock	weight_lb in the catalog; lots.rs; the shared quest host (phase, data gate, outcome application — not receipts); BaleQuest state and Arming; the leg override; the seizure; the opening and its resolution function — with no investigation content.	Headless: start, advance to Dayspring, watch the bolt found and Hugh seized. Quest-disabled prompts, snapshots and goldens byte-identical.
M1	The public weighing	Weighing, standards, the reader seam, heard_by. Bertran Hobbe's proved weights as a real item and his Lowmarket proving as a real event. The whole lawful lane: reweigh, amend, pay duty, settle the notice.	GO/NO-GO. If a lawful reweigh cannot be staged out of existing schedules and speech without a quest-only verb, the design is wrong and should be reworked here.
M2	The evidence surface	bale.json: the seven traces as authored facts with sealed holder sets and per-holder phrasings — no Rust. Arming wired. The three examinations (seal, cord, beam) as dispositions: who will say what they know. The scoped warehouse key. Deterministic fake-backend answers.	Every trace reachable headless; FactSource and pressed_by appear in no rendered string.
M3	The four lanes	Social (three-in-one-room, Skell buying the bolt, the six households), covert (removing the bolt, re-laying the cord, forging a seal, altering the custody copy), predatory (selling the truth, taking the key). Notice escalation and custody integration, including the player being taken.	Four materially different routes reach a cleared Hugh; correctness holds with the Night Office disabled.
M4	Opening and aftermath	The full public opening presentation; five outcome packages applied to the live world; follow-up lines that make the new state legible with no ending card.	All five opening branches reachable from a scripted run; the aftermath is visible without opening a menu.
M5	Content, polish, ship	All six loom households, every failure row and recovery route, prompt tuning, a full golden re-bless, drive scripts per lane.	Every failure-matrix row has a scripted run that continues to a resolution.

Dependencies and decisions needed before M0

Item	State	Handling
Knowledge and rumour	features/knowledge_and_rumor/ — SPEC ONLY	HARD PREREQUISITE. Owns Fact/holds(), the what_you_know block, pollen propagation and garbling, player receipts, the journal and World::arm_actor. Decided 2026-08-30 to ship both halves as one feature. Its own M4 is an open-ended perceptibility tuning pass, and three quests are behind it.
Shared quest scaffolding	Does not exist; three specs now each propose their own	Land quest.rs in M0 as a phase, a data gate and outcome application. Smaller than it was: receipts and the casebook moved to the knowledge layer, so it no longer commits the sibling specs to much.
Keys and locked places	features/keys_and_locked_places.md — SPEC ONLY	M2 ships one authored key item opening exactly one door (the bonded warehouse) with a hard-coded reach check. Honest, but a second half-implementation of a spec'd feature — if the keys feature is close, do it first. The quest must not block on it.
settle_notice on a false accusation	The ladder settles by returning `taken` or by the wronged party's say-so	Neither fits "the accused did not do it". M1 must decide whether amendment discharges a notice or whether a new discharge reason is needed. This is a change to shared law code, not quest code.
General credit path	Only round::try_purchase moves sparks	Duty payment and surety go through a narrow quest-owned settlement calling the same code. Third feature in a row to hit this (chalking, crowd knob). Worth fixing properly rather than working around a fourth time.
Memory / goal injection	state.memories and state.goal are seed-only plus LLM-editable; no sim-side setter (actions.rs:2325, 2603)	World::arm_actor lands with the knowledge layer. A seed, not an override — the actor's own set_goal/forget must win afterwards. Used for two characters here; if a fact can be a fact, it should be one.
Persistence	No save exists	The arc is deliberately a single session (~1.5 game days). Outcomes apply to the live world and are lost on quit, as everything else is.

Two content risks

- **Two people are called Clemence** — Clemence Hobbe the weaver (whose bolt it is) and Clemence Crake of the Tallage (Renn's mother, who owns the warehouse). Both are load-bearing and both are shipped. No rendered string may ever contain a bare “Clemence”; a test asserts it over every string the feature adds. An officer or an LLM confusing them mid-investigation is desirable; the author being confused is not.
- **The smoothest route (persuading Ewart Skell to buy the bolt) quietly destroys the independence of the person the quest is about, and nothing tells the player. That is a deliberate design position, not an oversight** — but it is the one thing in the feature that should be confirmed in playtest rather than assumed.

Acceptance criteria (abridged)

- **Determinism** — cargo test -p cathedral-sim passes, all offline. With bale.json absent, golden prompts are byte-identical. No HashMap order reaches a snapshot, prompt or golden.
- **The arithmetic property** — a test removes one bolt from a sealed lot and asserts the discrepancy goes to zero with no other state touched. true_weight_lb is never cached.
- **Coverage** — all five opening branches reachable headless; at least four materially different routes clear Hugh, at least two of them exposing neither Clemence Hobbe nor Renn; every failure-matrix row continues to a resolution rather than a dead end.
- **Cost** — the stage cap and the single in-flight cognition slot are unchanged; the quest adds no LLM lane. PublicSnapshot size unchanged (the canary still passes). Outcome unaffected at CATHEDRAL_EXTRA_NPCS=2000.

Vertical slice

```
cargo run -p cathedral-backends --bin cathedral-headless -- \
  --fake --quest bale --start-office highwick \
  --seconds-per-day 600 --watch-clock 0.5
```

Expected transcript: the gate stop and turnaround; a notice raised against rbrde; a weighing staged at the Tallage beam on Bertran Hobbe's proved weights before Odo Trask; an amendment lodged; forty sparks of duty paid; the notice settled; the Dayspring opening reduced to a formality; and three lines later, Ewart Skell at the Draper's Reach saying what he now pays by the ell.